

Henry Alexander Cavanagh

Curriculum Vitae

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Education

- 2018 - 2022 **PhD, Deep Learning & Biophysics**, Imperial College London.
Industry-sponsored PhD (Syngenta). Developed hybrid deep learning and biophysical models for solving problems in drug discovery and immunology.
Supervisor: Professor Robert Endres.
Thesis title: Computing Interpretable Representations of Cell Morphodynamics.
- Two major lead-author international research projects.
 - AI-driven drug discovery paper in Nature Communications. Collaborated with experimentalists at Syngenta in the UK and Switzerland. Cavanagh, Henry, et al. "Physics-informed deep learning characterizes morphodynamics of Asian soybean rust disease." Nature communications 12.1 (2021): 1-9.
 - 3D T cell migration paper (publication pending). Collaborated with immunologists at UNSW Sydney. Cavanagh, Henry, et al. "T cell morphodynamics reveal periodic shape oscillations in 3D migration".
 - Gave talks at international physics conferences (APS 2021 & 2022).
- 2014 - 2018 **MSci, Physics with Theoretical Physics**, Imperial College London.
- Top 2% in cohort in the year 3 Comprehensive examinations, testing problem solving abilities across the breadth of fundamental physics and mathematics.
 - Final year project: Deciphering momentum diffusivity behaviours within tropical cyclones using data from reconnaissance flights.
 - Society committee positions: Physics Society social secretary, Snowsports secretary.
- 2012 - 2014 **A levels**, St Edward's School, Cheltenham.
Mathematics A*, Further Mathematics A*, Physics A*, Chemistry A.
- 2010 - 2012 **GCSEs**, St Edward's School, Cheltenham.
10 A* grades, and the top grade in FSMQ Additional Mathematics.

Professional Experience

- 2018 - 2022 **Syngenta**, PhD-Industry Partnership.
International collaboration as described in Education. We worked together to design and interpret experiments, and developed the modelling framework in accordance with the data regimes of industry. A great experience for understanding the utility of our novel methods through a commercial lens.
- 2020 **Worldcoin**, Field Testing.
A moonshot Silicon Valley based cryptocurrency startup incorporating biometrics in order to give everyone on Earth an equal initial stake, now valued at over \$1 billion. Assisted with field testing of the hardware and concept in London.
- 2019 **Imperial College London**, Graduate Teaching Assistant.
- Co-supervised and mentored two final year undergraduate projects (cell morphogenesis and pandemic dynamics).
 - Produced and delivered a deep learning lecture and problem set for Bioinformatics MSc students.
 - Teaching assistant for the final year undergraduate Integrative Systems Biology course.

- 2019 **Linklaters**, Legal technology competition finalist.
Submitted an essay titled 'AI and the Law: A Paradigm Shift' to magic circle law firm Linklaters. Consequently invited to a day developing legal tech ideas with the innovations team.
- 2018 - 2019 **The Student Tutors Group**, Private tutor.
Provided mathematics and physics private tuition across London.
- 2016 **Sandtable**, Summer intern.
An innovative data science company using agent-based models for predictive analytics. Contributions included identifying model shortcomings, sourcing and manipulating data for progression into markets such as the US, Thailand and Eastern Europe, and creating data visualisations in Python.

Skills & Achievements

- Project management and leadership skills from teaching and project supervision roles during my PhD.
- Technical skills:
 - Scientific computing skills across a range of Python libraries, including Tensorflow and Pytorch.
 - Expertise in a range of quantitative disciplines across mathematics, physics, computer science and biology.
- Two-time winner of the All Round Excellence cup at school, and recipient of the All Round scholarship to sixth form. Spanish and Latin cups twice each, along with Academic awards in sixth form.
- Charitable work including volunteering at Riding for the Disabled (RDA) and a development expedition to Borneo.

Specific Strengths

Interdisciplinary vision | creativity | communication skills | big-picture thinking | energy and focus.

Interests

I throw myself into anything that involves adventure and love to go trail running, skiing, downhill biking and backpacking. At Imperial College, I competed on the rowing and snowsports teams, and I was on the debating, public speaking, rugby and cricket teams in high school. I find daily meditation enormously powerful, and hope to go on a 10 day silent retreat in the coming years. Finally, I love reading literature that blurs the lines between disciplines like economics, philosophy and physics, and run an online book club exploring these themes.

References

References for all educational and professional experiences are available upon request.